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Systems and methods for installation of solar panel assemblies

Abstract

A method including loading solar panel assemblies onto a vehicle, transporting the solar panel assemblies on the vehicle to a pile row of a solar project installation site, and individually unloading the solar panel assemblies along the pile row at the installation site to facilitate installation of the solar panel assemblies on the pile row. Other embodiments are described.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of PCT Application No. PCT/US2022/043547, filed Sep. 14, 2022. The content of the above-identified application is incorporated by reference herein in its entirety.

TECHNICAL FIELD
(1) The present disclosure is directed to systems and methods for installation of solar panel assemblies.
BACKGROUND
(2) Solar panels are often used as alternative energy resources. Typically, a large number of solar panels are used in a solar project to maintain sufficient energy supply scale to be suitable for an electric utility. However, installation of solar panels is a labor intensive process that involves significant labor and equipment costs, and takes a significant amount of time to properly install to ensure the solar panels operate efficiently in a durable configuration for the contracted life-cycle of the solar project.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) To facilitate further description of the embodiments, the following drawings are provided, in which like references are intended to refer to like or corresponding parts, and in which:
(2) FIG. **1** illustrates a top perspective view of an exemplary cassette that is empty (not holding solar panels assemblies), according to certain embodiments;
(3) FIG. **2** illustrates a bottom view of the cassette of FIG. **1**, where the cassette is internally driven;
(4) FIG. **3** illustrates a top perspective view of the cassette of FIG. **1** filled with solar panel assemblies;
(5) FIG. **4** illustrates a top view of the cassette of FIG. **1** filled with solar panel assemblies;
(6) FIG. **5**A illustrates a top plan view of an exemplary mobile solar panel assembly facility, according to certain embodiments;
(7) FIG. **5**B illustrates a top perspective view of the first portion of the mobile solar panel assembly facility of FIG. **5**A;
(8) FIG. **5**C illustrates a top plan view of a second portion of the mobile solar panel assembly facility of FIG. **5**A;
(9) FIG. **5**D illustrates a top perspective view of the second portion of the mobile solar panel assembly facility of FIG. **5**A;
(10) FIG. **5**E illustrates a top perspective view of a third portion of the mobile solar panel assembly facility of FIG. **5**A;
(11) FIG. **5**F illustrates a side elevational view of the third portion of the mobile solar panel assembly facility of FIG. **5**A;
(12) FIG. **5**G illustrates a side elevational view of a close-up of a portion of the jib boom and the conveyor, as shown in FIGS. **5**E-**5**F;
(13) FIG. **6** illustrates a top, front, passenger-side perspective view an exemplary solar panel assembly installation vehicle, according to certain embodiments;

(14) FIG. **7** illustrates a top, front, driver-side perspective view of the solar panel assembly installation vehicle of FIG. **6**;
(15) FIG. **8** illustrates a front elevational view of the solar panel assembly installation vehicle of FIG. **6**;
(16) FIG. **9** illustrates a top plan view of the solar panel assembly installation vehicle of FIG. **6**;
(17) FIG. **10** illustrates a top plan view of an exemplary solar project installation site, according to certain embodiments;
(18) FIG. **11** illustrates a top perspective view of the first assembly station of FIG. **5**A;
(19) FIG. **12** illustrates a top perspective view the first assembly station of FIG. **5**A without the structure of FIG. **11**, showing a jig;
(20) FIG. **13** illustrates a portion of the torque tube support structure and the hard stops of the jig of FIG. **12**;
(21) FIG. **14** illustrates a panel aligner of the jig of FIG. **12**;
(22) FIG. **15** illustrates a compressing actuator of the jig of FIG. **12**;
(23) FIG. **16** illustrates a lengthwise cross-section of the spacer rod of FIG. **13**;
(24) FIG. **17** illustrates a perspective view of a strip station, which can be included in another embodiment of the mobile solar panel assembly facility of FIG. **5**A;
(25) FIG. **18** illustrates a side view of the rotating drum of FIG. **17**;
(26) FIG. **19** illustrates a perspective view of the rotating drum and de-stacker of FIG. **17**;
(27) FIG. **20** illustrates an end view of the rotating drum and de-stacker of FIG. **17**;
(28) FIG. **21** illustrates a method of loading solar panel assemblies on a cassette, according to another embodiment;
(29) FIG. **22** illustrates a method of installing solar panel assemblies, according to another embodiment;
(30) FIG. **23** illustrates a perspective view of an assembly station, according to another embodiment;
(31) FIG. **24** illustrates an internal view of the automatic bracket loader of FIG. **23**; and
(32) FIG. **25** illustrates a method of assembling a solar panel assembly, according to another embodiment.
(33) For simplicity and clarity of illustration, the drawing figures illustrate the general manner of construction, and descriptions and details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the present disclosure. Additionally, elements in the drawing figures are not necessarily drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help improve understanding of embodiments of the present disclosure. The same reference numerals in different figures denote the same elements.
(34) The terms "first," "second," "third," "fourth," and the like in the description and in the claims, if any, are used for distinguishing between similar elements and not necessarily for describing a particular sequential or chronological order. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments described herein are, for example, capable of operation in sequences other than those illustrated or otherwise described herein. Furthermore, the terms "include," and "have," and any variations thereof, are intended to cover a non-exclusive inclusion, such that a process, method, system, article, device, or apparatus that comprises a list of elements is not necessarily limited to those elements, but may include other elements not expressly listed or inherent to such process, method, system, article, device, or apparatus.
(35) The terms "left," "right," "front," "back," "top," "bottom," "over," "under," and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the apparatus,

methods, and/or articles of manufacture described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.
(36) The terms “couple,” “coupled,” “couples,” “coupling,” and the like should be broadly understood and refer to connecting two or more elements mechanically and/or otherwise. Coupling may be for any length of time, e.g., permanent or semi-permanent or only for an instant. The absence of the word “removably,” “removable,” and the like near the word “coupled,” and the like does not mean that the coupling, etc. in question is or is not removable.
(37) As defined herein, two or more elements are “integral” if they are comprised of the same piece of material. As defined herein, two or more elements are “non-integral” if each is comprised of a different piece of material.
(38) As defined herein, “approximately” can, in some embodiments, mean within plus or minus ten percent of the stated value. In other embodiments, “approximately” can mean within plus or minus five percent of the stated value. In further embodiments, “approximately” can mean within plus or minus three percent of the stated value. In yet other embodiments, “approximately” can mean within plus or minus one percent of the stated value.
DESCRIPTION OF EXAMPLES OF EMBODIMENTS
(39) The present disclosure relates to systems and methods for installation of photovoltaic (PV) solar panels. Conventional approaches for utility-scale installation of solar panels at solar projects (e.g., 5 megawatts (MW) to 500 MW or more) typically involve single-axis tracker systems that sit on a structural shape piling system. The tracker systems generally include a torque tube, brackets that affix the torque tube to the module frames, bushings that affix to the pile caps, and the wire harness assembly that daisy chains output of the module rows. In conventional approaches, each of these components is pre-staged by teams on vehicles (trucks) at intervals in the field and subsequently hand assembled. Installing the system on existing piles involves a minimum of four steps, including tracker installation, alignment, module installation on the tracker system, and finally installation of the wire harnesses. There are other support tasks like garbage cleanup, pallet pickup, and an array of work assist equipment like forklifts, in addition to dealing with material shortages or damage that slows progress of the installation.
(40) Difficulties with these conventional approaches include that the work is manual, which presents problems and costs for labor acquisition, retention, training, and consistency of product quality. These factors can affect project cost and risk for a contractor, especially related to end-dates. This risk is generally priced into a contractor offering, absent a better way to manage these factors, but constrains any effort to compress installation costs of utility scale solar thereby limiting solar installed cost compression which is desirable for sustainability as a long-term power alternative.
(41) Embodiments disclosed herein can include the ability to improve cycle time and quality because the process is repeatable and far less dependent on impromptu team performance of varied work crews. Embodiments disclosed herein can reduce turnover and improve efficiency of the aggregate workforce associated with the installation of solar panel assemblies. In particular, the four (or more) serial field activities that exist with the conventional methods can become a single installation activity, with elimination of the field prestaging and material handling. Assemblies can be fabricated in precision repeatable jig and wire harnesses installed to repeatable visual indications making all assemblies consistent. The complete assembly can be loaded onto a cassette, and handled as a unit. In addition, cycle time per assembly can be faster and consistent using the embodiments disclosed herein. These embodiments can allow improvement to be made to quality and predictability to production end-dates, which are not easily achieved with the conventional methods.
(42) Overall, embodiments disclosed herein can improve the labor-intensive nature of solar installation, reduce installation costs, and/or improve efficiencies associated with the assembly and installation of solar panels. Embodiments disclosed herein can involve assembly of solar panel

assemblies at a mobile facility in the field at a solar project installation site, with internal transport and installation of the solar panel assemblies within the solar project installation site at various locations along pile roles within the solar project installation site. These embodiments can provide a vast improvement over conventional systems and methods which involve a plurality of steps and coordination among various parties to assemble and install the tracker systems in the field. Accordingly, embodiments disclosed herein can improve on conventional systems by reducing the time involved to install a solar panel assembly, reduce labor, and/or provide consistency to installation of solar panel assemblies.
(43) Turning to the figures, FIG. **1** illustrates a top perspective view of a cassette **100** that is empty (not holding solar panel assemblies, e.g., solar panel assemblies **300** in FIG. **3**), according to certain embodiments. The cassette **100** is merely exemplary, and embodiments of the cassette are not limited to the embodiment shown and presented herein. The cassette can be employed in many different embodiments or examples not specifically depicted or described herein. The cassette **100** of the illustrated example includes a base **102** and stanchion rows **104**. In some embodiments, a quantity of the stanchion rows **104** is at least eight. For example, as shown in FIG. **1**, the cassette **100** can include twelve stanchion rows, namely stanchion rows **131-142**. The stanchion rows **104** can each extend lengthwise along the cassette **100**. The stanchion rows **104** can be configured to hold solar panel assemblies (e.g., solar panel assemblies **300** in FIG. **3**). In some embodiments, each of the stanchion rows **104** is configured to hold a torque tube of a solar panel assembly.
(44) In the illustrated embodiment, each of the stanchion rows **104** can include stanchions **106**. In many embodiments, the stanchions **106** can be positioned in the cassette across the stanchion rows **104** in columns, such as columns **151-156**. In some embodiments, a quantity of the stanchions **106** within each of the stanchion rows **104** is at least three, to support the solar panel assembly at the ends and in at least one spot between the ends to limit the solar panel assembly from flexing along the torque tube, as the length of the solar panel assembly in some embodiments can be approximately 40 feet (12.19 meters (m)). For example, as shown in FIG. **1**, stanchion row **131** can include six stanchions, one at each of columns **151-156**. As depicted in FIG. **1**, the stanchions **106** in stanchion rows **131-141** in columns **151** and **153-156** are in upright positions, and the stanchions **106** in column **152** and the stanchions **106** in stanchion row **142** are in lowered positions. In some embodiments, each of the stanchions **106** of each of the stanchion rows **104** can include a saddle **108** configured to receive the torque tube of the solar panel assembly held by the stanchion row **104**. In some embodiments, the saddle **108** is positioned external to the tube **110**.
(45) In some embodiments, the saddle **108** is adjustable with respect to the stanchion **106** for one or more of the stanchions with each of the stanchion rows **104**. For example, in the illustrated embodiment, each of the stanchions **106** can include a tube **110** and a rod **112** positioned in the tube **110**. In some embodiments, the saddle **108** of the stanchion **106** is coupled to the rod **112**. In the illustrated embodiment, the rod **112** is coupled to an actuator **114** on the base **102**. In some embodiments, the actuator **114** can operate the rod **112** to extend the saddle **108** away from respective tube **110** and raise the position of saddle **108** above base **102**, thereby lifting, or supporting at a higher position, a torque tube of the corresponding solar panel assembly. In some embodiments, the actuator **114** can be operated by a foot of a worker to adjust the vertical position of the saddle **108**. In some embodiments, the rod **112** can be supported by a spring, such that the spring can bias the rod **112** and, thus, the saddle **108** outward from the tube **110**.
(46) In some embodiments, the base **102** can include one or more recesses **116** corresponding to each of the stanchion rows **104**. In some embodiments, the recesses **116** are configured to receive a portion (e.g., a bottom portion) of one or more solar panels of the respective solar panel assembly that are held by the stanchion rows **104** when the torque tube of the solar panel assembly is positioned in the saddles **108** of the stanchions **106** of the stanchion rows **104**.
(47) In some embodiments, the base **102** can include one or more apertures **118** configured to receive one or more tools or conveyors of a work vehicle. For example, the apertures **118** can

receive a tool of a work vehicle to maneuver the cassette **100** from one location to another.
(48) In some embodiments, the solar panel assemblies held by the stanchion rows **104** in the cassette **100** are configured to be installed in a solar project and extend across a full block row of a solar project. A full block row generally can have a variable number of solar panel assemblies, typically between six and twelve (e.g., twelve torque tubes, with the associated solar panels affixed to each of the torque tubes) and an aggregate of approximately 144 solar panels, though block rows usually have approximately 100 solar panels. For example, in some embodiments, the solar panel assemblies configured to be held by the stanchion rows **104** each can include up to twelve solar panels.
(49) In some embodiments, the stanchion rows **104** are configured to hold the solar panel assemblies when the lengths of torque tubes of the solar panel assemblies are different and when the lengths of the torque tubes are the same. For example, the torque tubes can be of variable length, depending on specifications for installation of the solar panel assemblies at particular locations of the solar project. In some cases, a solar panel assembly can contain between five and twelve solar panels and the torque tube of the solar panel assembly can extend merely halfway from a first end of the base **102** to a center point of the base **102**. Accordingly, a portion of the stanchions **106** can receive the torque tube, while other stanchions **106** in another stanchion row **104** can hold a solar panel assembly with 12 solar panels and a torque tube that across the length of the base **102** from column **151** to column **156**, for example. In some embodiments, an extender rod (e.g., **648**, as shown in FIGS. **6** and **9**) can be used to span the rest of the stanchion row **104** if the torque tube does not extend across the entire stanchion row. The extender rod can be used to provide uniform overall length of the solar panel assembly while the solar panel assembly is being loaded onto the cassette, while the solar panel assembly is being held within the cassette, and/or while the solar panel assembly is being removed from the cassette.
(50) In some embodiments, one or more of the stanchions **106** of each of the stanchion rows **104** can be configured to rotate back-and-forth between an upright position (e.g., the positions of the stanchions **106** in column **152**) and a lowered position (e.g., the positions of the stanchions in columns **151** and **153-156**). In some embodiments, the stanchions **106** can be spring-biased in the upright position, such that the stanchion **106** can rotate from the lowered position to the upright position when a worker releases a hook or latch (not shown) that is holding the stanchion **106** in the lowered position. When the worker desires certain stanchions **106** to be lowered, such as when the solar panel assembly has been removed from the stanchions **106** of a certain stanchion row **104**, the worker can rotate the stanchions **106** from the upright position to the lowered position.
(51) Turning ahead in the drawings, FIG. **2** illustrates a bottom view of the cassette **100**. In some embodiments, the base **102** can include a drive system including a first set of wheels **200** and a second set of wheels **202**. In the illustrated embodiment, the first set of wheels **200** is positioned in a first direction, and the second set of wheels **202** is positioned in a second direction different from the first direction. For example, as shown in FIG. **2**, the first set of wheels **200** can be oriented to move the cassette **100** in a widthwise direction, and the second set of wheels **202** can be oriented to move the cassette **100** in a lengthwise direction. In some embodiments, the drive system of base **102** can include drive motors (e.g., **204**) configured to drive the first set of wheels **200** and/or the second sets of wheels **202**. In some embodiments, the base **102** can include a battery **206** configured to power the drive motors **204**. The drive system can be used to move the cassette **100**. In other embodiments, the cassette **100** does not include a drive system, and various conveyors can be used to move the cassette **100**.
(52) Turning ahead in the drawings, FIG. **3** illustrates the cassette **100** filled with solar panel assemblies **300**, such as solar panel assembly **301**. FIG. **4** illustrates a top view of the cassette **100** filled with the solar panel assemblies **300**. The stanchion rows **104** can each hold a separate solar panel assembly (e.g., **301**). As shown in FIGS. **3-4**, the cassette **100** can hold twelve solar panel assemblies (e.g., **300**). In the illustrated embodiment, the solar panel assemblies **300** include torque

tubes, such as a torque tube **302** of solar panel assembly **301**. In the illustrated embodiment, the torque tubes (e.g., **302**) are positioned in saddles **108** of stanchions **106** within each of the stanchion rows **104**. For example, as shown in FIG. **3**, torque tube **302** of solar panel assembly **301** is seated within saddles **108** of stanchions **106** along stanchion row **131**.
(53) As described in more detail below, solar panels, such as solar panel **304**, can be affixed to torque tube **302** of solar panel assembly **301**. In the embodiment shown, each of the solar panel assemblies **300** includes twelve solar panels, such that the cassette **100** collectively holds 144 solar panels. Other configurations of solar panel assemblies with different numbers of solar panels (and even varying numbers of solar panels across stanchion rows **104**) can be held by the cassette **100**. As shown in FIGS. **3-4**, when the solar panel assemblies **300** are held by the cassette **100**, the solar panels within the solar panel assemblies can be positioned in a substantially vertical position. In many cases, each of the solar panels is at approximately 78 inches (2 m) long (e.g., tall in the upright position) and approximately 39 inches (1 m) wide. As assembled, each of the solar panels assemblies can weigh approximately 1,200 pounds (544.3 kilograms (kg)).
(54) Turning ahead in the drawings, FIG. **5**A illustrates a top plan view of a first portion of an exemplary mobile solar panel assembly facility **500**, according to certain embodiments. FIG. **5**B illustrates a top perspective view of the first portion of the mobile solar panel assembly facility **500**. The mobile solar panel assembly facility **500** is merely exemplary, and embodiments of the mobile solar panel assembly facility are not limited to the embodiment shown and presented herein. The mobile solar panel assembly facility can be employed in many different embodiments or examples not specifically depicted or described herein. In the illustrated embodiment, the mobile solar panel assembly facility **500** includes an unload queue **504**, a load queue **506**, and one or more assembly stations (e.g., **526**, **528**). Unload queue **504** can be used to unload empty cassettes (e.g., no solar panel assemblies in the cassettes), such as empty cassettes **531-532**, from vehicles, such as a first vehicle (not shown) in a truck unloading spot **502**. Load queue **506** can be used to load filled cassettes (e.g., filled with solar panel assemblies in the cassettes), such as filled cassettes **535**, onto vehicles, such as a second vehicle (not shown) in a truck loading spot **508**. Additionally, the load queue **506** can be used to fill empty cassettes (e.g., **533**, **534**) from the assembly stations **526**, **528**. Cassettes **531-536** can be similar or identical to cassette **100** (FIG. **1**), and various elements of cassettes **531-536** can be similar or identical to various elements of cassette **100** (FIG. **1**). The first vehicle (e.g., a vehicle in truck unloading spot **502**) and/or the second vehicle (e.g. a vehicle in truck loading spot **508**) can be similar or identical to solar panel assembly installation vehicle **600**, as shown in FIGS. **6-9** and described below.
(55) In some embodiments, the first vehicle is different from the second vehicle, such that the first vehicle can be unloaded in truck unloading spot **502** while the second vehicle is loaded in truck loading spot **508**. Once the second vehicle in truck loading spot **508** is loaded and drives forward and away from the truck loading spot to an installation site, and the first vehicle has unloaded an empty cassette (e.g., **531-532**), the first vehicle can proceed from the truck unloading spot to the truck loading spot to be loaded with a filled cassette (e.g., **535**). In some embodiments, three or more vehicles can be used, such that concurrently one is being unloaded at unload queue **504**, one is being loaded at load queue **506**, and one or more is away at an installation site installing solar panel assemblies, and the three or more trucks rotate progressively through these actions.
(56) In many embodiments, the unload queue **504** and the load queue **506** of the mobile solar panel assembly facility **500** can be positioned on the top of a set of flatbeds, such as 8 feet (2.4 m) wide by 40 feet (12.2 m) long flatbeds (e.g., **541**, **542**). For example, as shown in FIGS. **5**A-**5**B, a surface **540** for the unload queue **504** and load queue **406** is formed by using 17 of these flatbeds positioned closely side-by-side next to each other from the flatbed **541** to the flatbed **542**. The surface **540** can be mobile, as the flatbeds are on wheels and can be moved by trucks to various different positions. Similarly, the assembly stations **526**, **528** can each be on custom enclosed vehicles and/or the top of flatbeds, such as 10 feet (3.0 m) wide by 48 feet (14.6 m) long flatbeds,

which extend lengthwise away from the side of flatbed **542** at the end of the surface **540** proximate to load queue **506**. In some embodiments, the mobile solar panel assembly facility **500** can be comprised of a combination of a ground surface, one or more flatbeds of a mobile vehicle, and/or one or more surfaces of a platform structure (e.g., lightweight platforms), instead of all flatbeds. For example, the unload queue **504** can be positioned on a platform structure, and the load queue **506** can be positioned on a flatbed. Embodiments of the assembly stations **526**, **528** are further described in detail below in connection with FIGS. **11-20**.

(57) During operation, a cassette (e.g., **531**) is transferred from the first vehicle in truck unloading spot **502** to the unload queue **504**. The unload queue **504** can include one or more cassettes (e.g., **531-532**), which can be used to supply empty cassettes to the load queue **506**. In some embodiments, the cassettes (e.g., **531-532**) can include drive systems, as shown in FIG. **2** and described above, which can be used to move the cassettes within the unload queue **504**. In other embodiments, as shown in FIGS. **5**A-**5**B, the unload queue **504** can include an unload conveyor **514** configured to move the cassette (e.g., **531-532**) within the unload queue **504**.

(58) In some embodiments, such as in the embodiments shown in FIGS. **5**A-**5**B, when a cassette (e.g., **536**) reaches a transfer conveyor **518**, the transfer conveyor **518** can transfer the cassette (e.g., **536**) from the unload queue **504** to the load queue **506**. In some embodiments, the load queue **506** can include a load conveyor **516** configured to move the cassettes within the load queue **506**. In other embodiments, the cassette includes the drive motors **204** (FIG. **2**) to transfer the cassette from the first vehicle in truck unloading spot **502** to the unload queue **504**, from the unload queue **504** to the load queue **506**, and from the load queue **506** to the second vehicle in truck loading spot **508**.

(59) While the cassette is in a fill position of the load queue **506** (such as the positions of cassettes **533-534** shown in FIGS. **5**A-**5**B), solar panel assemblies **300** (FIG. **3**) can be loaded into the slots of the cassette (e.g., **533**, **534**) from the assembly stations **526**, **528**, respectively. As discussed above, the cassette can include slots (e.g., stanchion rows **104** (FIGS. **1**, **3-4**) to receive and hold the solar panel assemblies (e.g., **300** (FIGS. **3-4**)). Once the slots of the cassette are filled with solar panel assemblies, the filled cassettes (e.g., **535**) can be transferred from the load queue **506** to the second vehicle in truck loading spot **508** to transfer the solar panel assemblies to an installation site.

(60) In some embodiments, loading the solar panel assemblies into the slots of the cassette (e.g., **533**, **534**) at the fill position can involve moving a solar panel assembly from the assembly station (e.g., **526**, **528**) at which it was assembled to the cassette. This solar panel assembly can be moved from the assembly station (e.g., **526**, **528**) to the cassette (e.g., **533**, **534**) at the fill position using conveyors, overhead transporters, and/or other suitable mechanisms. In some embodiments, loading the solar panel assemblies into the slots of the cassette (e.g., **533**, **534**) at the fill position can include, for each one of the solar panel assemblies, indexing a position of the cassette (**533**, **534**) to align a respective next empty one of the slots with an assembling position **512** of the mobile solar panel assembly facility to facilitate loading of each one of the solar panel assemblies from the assembling position **512** into the respective next empty one of the slots. For example, once a solar panel assembly **551** has been transferred from assembly station **526** onto cassette **533** and once solar panel assembly **552** has been transferred from assembly station **528** onto cassette **534**, the cassettes **533** and **534** can be moved approximately 12 inches along the load queue **506** to position the next slots of the cassettes at the assembling positions **512** of assembly stations **526**, **528**. In some embodiments, loading the solar panel assemblies (e.g., **300**) into the slots of the cassette (e.g., **533**, **534**) can include, for each one of the solar panel assemblies, raising stanchions (e.g., **106** (FIG. **1**)) at the respective next empty one of the slots to hold the solar panel assemblies, so that the slot can be ready to receive the next solar panel assembly from the assembly station (e.g., **526**, **528**).

(61) In some embodiments, loading the solar panel assemblies (e.g., **551**, **552**) into the slots of the cassette (e.g., **533**, **534**) further can include, for each one of the solar panel assemblies, coupling a

respective wire harness to each one of the solar panel assemblies. For example, once a solar panel assembly (e.g., **551**) has been loaded into a slot of the cassette (e.g., **533**), a wire harness can be installed from a wire harness sorting station **510** of the mobile solar panel assembly facility **500**. In some embodiments, a solar panel assembly can be loaded onto the cassette (e.g., **533**, **534**) approximately every 60 seconds. In some embodiments, the wire harness can be installed prior to loading the solar panel assemblies (e.g., **551**, **553**) into the slots of the cassette (e.g., **533**, **534**).

(62) In some embodiments, the cassette can move in a first widthwise direction **520** within the unload queue **504**. In some embodiments, the cassette **100** can move in a second widthwise direction **522** within the load queue **506**. In some embodiments, the second widthwise direction is opposite the first widthwise direction. In some embodiments, the cassette **100** can move in a lengthwise direction **524** when being transferred from the unload queue **504** to the load queue **506**.

(63) In some embodiments, the solar panel assemblies are assembled at the mobile solar panel assembly facility **500**. In some embodiments, the assembly stations **526** and **528** assemble solar panel assemblies concurrently by receiving torque tubes from a torque tube sorting area **550** (shown in FIGS. **5**E-**5**F and described below) that extends widthwise off of assembly station **526**, and receiving solar panels from solar panel conveyors (shown in FIGS. **5**C-**5**D and described below) that extend lengthwise off of assembly stations **526** and **528**, respectively. The assembly stations **526** and **528** each can add bushings to the torque tube and affix the solar panels to the torque tube using brackets to form the solar panel assembly. In some embodiments, as described above, the wire harness is affixed as soon as the solar panel assembly is transferred to the cassette, while the next solar panel assembly to be loaded on the cassette is being assembled at the assembly station (e.g., **526**, **528**),

(64) Turning ahead in the drawings, FIG. **5**C illustrates a top plan view of a second portion of the mobile solar panel assembly facility **500**. FIG. **5**D illustrates a top perspective view of the second portion of the mobile solar panel assembly facility **500**. In the illustrated embodiments, the second portion of the mobile solar panel assembly facility **500** includes a first solar panel loading and sorting station **560** and a second solar panel loading and sorting station **580**. In many embodiments, first solar panel loading and sorting station **560** can extend off of the first assembly station **526** (FIGS. **5**A-**5**B), and second solar panel loading and sorting station **580** can extend off of the second assembly station **528** (FIGS. **5**A-**5**B). First solar panel loading and sorting station **560** can include a loading area **561**, a waste station **562**, a pallet conveyor **563**, a pallet stripping area **564**, and an accumulation area **566**. Second solar panel loading and sorting station **580** can include similar elements, such as a loading area **581**, a waste station **582**, a pallet conveyor **583**, a pallet stripping area **584**, and a sorting area **586**, which can be similar or identical to loading area **561**, waste station **562**, pallet conveyor **563**, pallet stripping area **564**, and accumulation area **566**, respectively.

(65) During operation, solar panel packages **570** are unloaded from a vehicle at the loading area **561** and positioned on the pallet conveyor **563**. The pallet conveyor **563** operates to move the solar panel packages **570** toward the pallet stripping area **564**. At the pallet stripping area **564**, the solar panel packages **570** are unpacked and the waste (e.g., **572**) is unloaded at the waste station **562**. In some embodiments, the solar panel packages **570** can include solar panels, cardboard, and pallets. In some embodiments, the solar panels **574** are removed from the solar panels packages **570**, and the solar panels **574** are moved to the accumulation area **566** while the packaging of the solar panel packages **570** are stripped and unloaded at to the pallet stripping area **564**. In some embodiments, a portion **576** of the solar panels **574** are fed into the accumulation area **566** based on a solar panel assembly specification. In some embodiments, portion **576** can be in a different orientation from solar panels **574**, which can be achieved using strip station **1700** (FIG. **17**, described below). For example, a solar panel assembly may be specified to include 8 solar panels. Accordingly, the portion **576** can represent 8 solar panels from the solar panels **574** from the solar panel package **570**. Based on the solar panel assembly specification, each solar panel from the portion **576** is loaded individually onto a conveyor **578** in the accumulation area **566**. Conveyor **578** can move the

solar panels along the accumulation area **566** to the first assembly station **526** (FIGS. **5**A-**5**B), where the solar panels can be aligned and coupled together (discussed in more detail below) on a torque tube, after which the solar panel assembly is loaded onto a cassette (e.g., **533** (FIGS. **5**A-**5**B)) and transferred to the installation site. Second solar panel loading and sorting station **580** can operate similar to first solar panel loading and sorting station **560**. In some embodiments, the solar panels **574** are fed into the accumulation area **566** to maintain the accumulation area **566** at a full capacity with solar panels. For example, each solar panel from the solar panels **574** is loaded individually onto a conveyor **578** in the accumulation area **566**. Conveyor **578** can move the solar panels along the accumulation area **566** to the first assembly station **526** (FIGS. **5**A-**5**B), and additional solar panels **574** can be fed into the accumulation area **566** to fill the space of the solar panels **574** that have been positioned in first assembly station **526**.

(66) Turning ahead in the drawings, FIG. **5**E illustrates a top perspective view of a third portion of the mobile solar panel assembly facility **500**. FIG. **5**F illustrates a side elevational view of the third portion of the mobile solar panel assembly facility **500**. In the illustrated embodiments, the third portion of the mobile solar panel assembly facility includes the torque tube sorting area **550**. In some embodiments, the torque tube sorting area **550** can be positioned on the top of a set of flatbeds **553**, or in enclosed trailers. The torque tube sorting area **550** can include a set of conveyors **555** on each of the flatbeds **553**. For example, the set of conveyors **555** can include conveyors **591-594**. In some embodiments, the torque tube sorting area can include jib booms **556**, which can be used to move torque tubes **559** from a supply area **554** onto the set of conveyors **555**. Various ones of the torque tubes **559** can be placed on different ones of the conveyors (e.g., **591-594**) depending on the length of the torque tubes **559**, such that different conveyors (e.g., **591-594**) carry torque tubes (e.g., **559**) of different lengths. In some embodiments, torque tubes can be fit with a torque tube extension to ensure the torque tube is stable and can fit within the conveyors (e.g., **591-594**).

(67) In some embodiments, a conveyor **557** can the transfer the torque tubes **559** from the set of conveyors **555** to first assembly station **526**, and a conveyor **558** can transfer the torque tubes **559** from the set of conveyors **555** to the second assembly station **528**. In some embodiments, as shown in FIG. **5**F, the conveyor **558** can lift the torque tubes **559** at a first side of the first assembly station **526** and lower the torque tubes **559** at a second side of the first assembly station **526** to provide the torque tubes **559** to the second assembly station **528**.

(68) Turning ahead in the drawings, FIG. **5**G illustrates a side elevational view of a close-up of a portion of the jib boom **556** and the conveyor **591**, as shown in FIGS. **5**E-**5**F. As shown in FIG. **5**G, the jib boom **556** can include a retention element **595** that is attached to a wire **596**. In some embodiments, the wire **596** can be connected to the jib boom **556** using a hook **598**. In some embodiments, a load balancer **597** can be included between the hook **598** and the wire **596**. In a number of embodiments, the retention element **595** can be configured to retain and lift the torque tubes **559**, such as to place the torque tubes **559** on conveyor **591** or another one of the conveyors (e.g., **592-594** (FIGS. **5**E-**5**F)). In some embodiments, the torque tube sorting area can include a load balancer on a set of monorails, which can be used to move torque tubes **559** from a supply area **554** onto the set of conveyors **555**, or other suitable elements.

(69) Jumping ahead in the drawings, FIG. **10** illustrates a top plan view of an exemplary solar project site **1000**, according to certain embodiments. As shown in FIG. **10**, the solar project site **1000** can be next to a main road **1002** and a side road **1003**, with plant roads **1005** throughout the area of the solar project site **1000**. In some embodiments, the solar project site **1000** can include an administration building **1004** and a switchyard **1006**. In some embodiments, the solar project site **1000** can include an area that is sectioned into blocks, such as blocks **1011-1013**. In some embodiments, each block can include approximately 30 rows of solar panels, with each row having approximately 100 solar panels. For example, twelve solar panel assemblies can be installed on a row, and joined together to form a row of approximately 100 solar panels. In such a configuration, each block can be approximately 600 feet (182.9 m) by 400 feet (121.9 m), and can produce 1 MW.

In the example shown in FIG. **10**, there are 252 blocks. The blocks can be built in direction **1001**. The first 140 of these blocks can be designated as part of a first build area **1010**, and the second 112 of these blocks can be designated as part of a second build area **1020**. While solar panel assemblies are being installed in the blocks of the first build area **1010**, the mobile solar panel assembly facility **500** (FIGS. **5**A-**5**F) can be located at a block **1014** that is centrally located within the first build area **1010**. When the blocks of the first build area **1010** are completed, the mobile solar panel assembly facility **500** (FIGS. **5**A-**5**F) can be relocated to a block **1024** that is centrally located within the second build area **1020**. By relocating the mobile solar panel assembly facility **500** (FIGS. **5**A-**5**F) during the build of solar project site **1000**, the driving distance from the mobile solar panel assembly facility **500** to the installation site along the rows of the blocks can be limited. In some embodiments, block **1014** and/or block **1024** can be installed with solar panels after the mobile solar panel assembly facility **500** (FIGS. **5**A-**5**F) are moved away from those respective blocks. In other embodiments, the solar panel assemblies can be assembled outside of the project site envelope (complex) or fully away from the solar project site **1000**, such as at a non-mobile solar panel assembly facility, and transported to the solar project site **1000**.

(70) Turning back in the drawings, FIG. **6** illustrates a top, front, passenger-side perspective view of an exemplary solar panel assembly installation vehicle **600**, according to certain embodiments. FIG. **7** illustrates a top, front, driver-side perspective view of the solar panel assembly installation vehicle **600**. FIG. **8** illustrates a front elevational view of the solar panel assembly installation vehicle **600**. FIG. **9** illustrates a top plan view of a portion of the solar panel assembly installation vehicle **600** without a support structure **604** and an unloader **606**. In some embodiments, the solar panel assembly installation vehicle **600** includes a cab **601**, a vehicle truck bed **602**, the support structure **604**, and/or the unloader **606**. In some embodiments, the vehicle truck bed **602** includes a first end **608** and a second end **610**. In some embodiments, the support structure **604** includes (i) a first vertical portion **612** positioned at the first end **608** of the vehicle truck bed **602**, (ii) a second vertical portion **614** positioned at the second end **610** of the vehicle truck bed **602**, and (iii) a horizontal portion **616** that is coupled to the first vertical portion **612** and the second vertical portion **614**. In other embodiments, other configurations of support structure can be used, such as arches, domes, or other suitable support structures. In some embodiments, the unloader **606** is coupled to the horizontal portion **616** and configured to unload the solar panel assemblies **300** from the cassette **100** that is being carried on the vehicle truck bed **602**.

(71) In some embodiments, the unloader **606** can include telescoping monorails **618**. In other embodiments, other suitable unloaders can be used, such as jib booms or other suitable unloaders. In some embodiments, the telescoping monorails **618** include a first portion **620** coupled to the horizontal portion **616** and a second portion **622** coupled (e.g., slidably coupled) to the first portion **620**. In some embodiments, the second portion **622** is configured to move along a length of the first portion **620**.

(72) In some embodiments, the telescoping monorails **618** include a load balancer (not shown). In some embodiments, the telescoping monorails include a cable **626** having a first end adjacent to the second portion **622** and a second end. In some embodiments, the telescoping monorails **618** includes a retention tool **728** coupled to the second end of the cable **626**. In some embodiments, the retention tool **728** is configured to retain and lift the solar panel assemblies **300** individually.

(73) In some embodiments, the support structure **604** is removably attached to the solar panel assembly installation vehicle **600**. In some embodiments, the support structure **604** is configured to be removable from the solar panel assembly installation vehicle **600** to facilitate transport of the solar panel assembly installation vehicle **600** on public roads away from the solar project installation area.

(74) In some embodiments, the solar panel assembly installation vehicle **600** includes a conveyor (not shown) that is configured to move the cassette **100** on the vehicle truck bed **602**.

(75) In some embodiments, the first vertical portion **612** includes a first frame spacer **632**, the

second vertical portion **614** includes a second frame spacer **634**, and the horizontal portion **616** includes a frame **636**. In some embodiments, the horizontal portion **616** includes a cover that extends across the frame **636**, which can provide shade and/or cover from rain.

(76) In some embodiments, the solar panel assembly installation vehicle **600** includes axles (not shown). In some embodiments, each of the axles is driven and steerable. As such, the solar panel assembly installation vehicle **600** is able to crab steer into a position along a pile row at an installation site to facilitate the installation of the solar panel assemblies **300**. In some embodiments, the solar panel assembly installation vehicle **600** can include one or more side-slope levelers (not shown) to maintain center-of-gravity within prescribed safety margins given the load and trim as needed to keep work environment level and comfortable. The axles allow for platform stability, minimal turn radius, and load stability. In some embodiments, the solar panel assembly installation vehicle **600** is outfitted with commercially available GPS system with fore and aft sensors so a driver can stay on a predetermined line in relation to the pile row.

(77) In some embodiments, the solar panel assemblies **300** include the torque tube **302**, a set of solar panels **640** affixed to the torque tube **302** with brackets **642**, bushings **644** configured to attach the torque tube **302** to pile caps (e.g., **801**) at the top of piles (e.g., **701**, **702**), and a wire harness **646**.

(78) In some embodiments, the solar panel assemblies on a filled cassette (e.g., **100**) are removed from the cassette when the solar panel assembly installation vehicle **600** (onto which the filled cassette has been loaded) arrives at an installation site. In some embodiments, removing the solar panel assemblies **300** from the cassette **100** can include individually lifting each respective one of the solar panel assemblies **300** and transferring each respective one of the solar panel assemblies **300** in a first widthwise direction to a position above a respective installation position along a pile row (e.g., of piles **701**, **702**) of the installation site. In some embodiments, each respective one of the solar panel assemblies **300** is lowered onto respective pile caps (e.g., **801** (FIG. **8**, described below)) at the respective installation position.

(79) For example, as shown in FIGS. **7-9**, when the solar panel assembly installation vehicle **600** is positioned at an installation position along a pile row of piles (e.g., **701-702**), a solar panel assembly **710** of solar panel assemblies **300** that was lifted to be removed from the cassette **100** and swung outward by the unloader **606** to be positioned over the pile row, the solar panel assembly **710** can be lowered onto the pile caps (e.g., **801**) at the top of the piles (e.g., **701-702**), such that the bushings **644** (FIG. **6**) engage with the pile caps (e.g., **801**).

(80) In some embodiments, a coupler is installed on a first torque tube of a first solar panel assembly to couple a second torque tube of a second solar panel assembly to the first torque tube. For example, there can be couplers installed between each set of torque tubes on a block row. In some embodiments, the second solar panel assembly is moved in a lengthwise direction toward the first solar panel assembly to position the second torque tube within the coupler and couple the second torque tube to the first torque tube. In some embodiments, the solar panel assembly installation vehicle **600** is driven forward to a respective subsequent one of the respective installation positions along the pile row between removing each respective one of the solar panel assemblies **300**.

(81) In some embodiments, the cassette **100** is maneuvered on the solar panel assembly installation vehicle **600** in a second widthwise direction opposite the first widthwise direction to facilitate removing the solar panel assemblies **300** from the cassette **100**. For example, the cassette **100** can be moved in the direction of the passenger side of the solar panel assembly installation vehicle **600** up to approximately 24 inches (60.96 centimeters (cm)) to provide space on the vehicle truck bed **602** for workers to assist with handling the solar panel assemblies, such as attaching the retention tool **728** to the solar panel assemblies **300**. In some embodiments, the cassette **100** can include a drive system, such as shown in FIG. **2** and described above, which can be used to move the cassette on the vehicle truck bed **602**. In other embodiments, the vehicle truck bed **602** can include a

conveyor to move the cassette **100** on the vehicle truck bed **602**. In some embodiments, as shown in FIGS. **7** and **9**, there can be additional room at the front of the truck bed and the rear of the truck bed beyond each end of the cassette **100** to allow the workers space to assist with handling the solar panel assemblies **300**.

(82) FIG. **11** illustrates a top perspective view of the first assembly station **526**. The first assembly station **526** is merely exemplary, and embodiments of the first assembly station are not limited to the embodiment shown and presented herein. The first assembly station **526** can be employed in many different embodiments or examples not specifically depicted or described herein. In the illustrated embodiment, the first assembly station **526** includes a base **1100** and a structure **1102**. In the illustrated embodiment, the base **1100** is a flatbed of a semi-trailer. In some embodiments, the base **1100** can be a ground surface, a flatbed of a mobile vehicle, or a surface of a fixed structure. In the illustrated embodiment, the structure **1102** includes (i) a first vertical portion **1104** positioned at the first end **1106** of the base **1100**, (ii) a second vertical portion **1108** positioned at the second end **1110** of the base **1100**, and (iii) a horizontal portion **1112** that is coupled to the first vertical portion **1104** and the second vertical portion **1108**. In a number of embodiments, structure **1102** can include a front portion **1105** and/or a rear portion **1107**. In some embodiments, at least a portion of structure **1102** can be covered to provide shade and/or cover from rain.

(83) The first assembly station **526** can include a torque tube load rack **1114**, which can be coupled to the base **1100** and/or to structure **1102**. In some embodiments, the torque tube load rack **1114** can include a linear positioner, such as hard stops **1116** or an automatically controlled linear positioner, and stanchions **1118**. In some embodiments, the stanchions **1118** are hydraulic stanchions, which can be adjusted to various different heights using a foot-activated actuator. In some embodiments, the stanchions **1118** are in a fixed position and are manually or automatically adjusted based on the specifications of a solar panel assembly. The stanchions **1118** can be configured to receive a torque tube **1120**. In some embodiments, the stanchions **1118** can include saddles, and in some embodiments, the saddles can include rollers to allow the rollers to allow the torque tube **1120** to rotate within the saddles. The hard stops **1116** can be configurable to abut ends of the various different torque tubes (e.g., **1120**) at different positions. In some embodiments, a solar panel assembly specification can specify that a torque tube of a solar panel assembly intended for a certain installation position has a certain amount of overhang at the end of the solar panel assembly beyond the end of the solar panels in order to be coupled with a neighboring solar panel assembly. In various embodiments, the torque tube **1120** can be loaded into torque tube load rack **1114** through an opening in front portion **1105**. In some embodiments, the torque tube **1120** can be loaded from the first end **1106** via an adjacent accumulation area (e.g., **566** (FIG. **5**)).

(84) In some embodiments, one of the hard stops **1116** can be selected and utilized to abut an end of the torque tube **1120** to account for the overhang. In some embodiments, a subsequent solar panel assembly can involve another amount of overhang different from the previous overhang. Accordingly, another one of the hard stops **1116** can be utilized to abut an end of a subsequent torque tube to account for the overhang. In some embodiments, brackets **1122** and bushings **1124** can be coupled to the torque tube **1120** based on the specifications of the particular solar panel assembly. In some embodiments, the hard stops **1116** are operated manually by an operator, or the hard stops **1116** can be operated autonomously based on the specifications of a particular solar panel assembly (e.g., measurement of overhang from a prior/future torque tube). In some embodiments, the brackets **1122** are positioned on the torque tube **1120** based on respective locations of corresponding solar panels, and the bushings **1124** are positioned on the torque tube **1120** based on respective locations of corresponding pile caps at an installation site. In some embodiments, the positions of the brackets and/or the bushings can be specified in the solar panel assembly specification.

(85) Turning ahead in the drawings, FIG. **12** illustrates a top perspective view of the first assembly station **526** without the structure **1102**, showing a jig **1200**. The jig **1200** is merely exemplary, and

embodiments of the jig are not limited to embodiments presented herein. The jig **1200** can be employed in many different embodiments or examples not specifically depicted or described herein. In a number of embodiments, the jig **1200** can be used to affix solar panels **1204** to a torque tube **1206** to form a solar panel assembly, with the solar panels aligned and spaced according to specifications. For example, in some embodiments, up to twelve solar panels (e.g., **1204**) can be affixed to a torque tube (e.g., **1206**) using the jig. In the illustrated embodiment, the jig **1200** is illustrated coupled to the base **1100** behind the torque tube load rack **1114**. In some embodiments, the jig **1200** can include a track system **1201**, a torque tube support structure **1202**, a panel aligner **1400** (FIG. **14**, described below), a compressing actuator **1224**, and/or hard stops **1226**.
(86) In some embodiments, the torque tube support structure **1202** is configured to support the torque tube **1206** in a fixed position. For example, after a torque tube has been positioned on torque tube load rack **1114**, and brackets and bushings have been installed on the torque tube, the torque tube can be moved to torque tube support structure **1202** and placed in the fixed position. In some embodiments, the track system **1201** can be configured to support and to align the solar panels **1204** along the torque tube **1206** when the torque tube **1206** is supported in the fixed position on torque tube support structure **1202**, to allow the solar panels **1204** to be affixed to the torque tube **1206** to form a solar panel assembly.
(87) In some embodiments, the track system **1201** can include a first track **1208** coupled to the base **1100** and a second track **1210** coupled to the structure **1102** (e.g., the horizontal portion **1112** of the structure **1102**). In some embodiments, the first track **1208** can include a first conveyor **1212** that is configured to move the solar panels **1204** along a length of the first track **1208**. In some embodiments, the second track **1210** is configured to pivot in a first direction **1214** to position the solar panels **1204** in the first track **1208**. In some embodiments, the second track **1210** is configured to pivot in a second direction **1216** to remove the solar panels **1204** from the first track **1208**. In some embodiments, the second track **1210** is parallel with the first track **1208** when the solar panels **1204** are positioned in the first track **1208** and the second track **1210**. In some embodiments, the second track **1210** is operated away from the solar panels **1204** and toward the solar panels **1204** via an actuator **1218**. In some embodiments, the actuator **1218** operates the second track **1210** to assist in maintaining and releasing the solar panels **1204** from the first track **1208**. In some embodiments, the second track **1210** includes a conveyor that is configured to move the solar panels **1204** along a length of the first track **1208**. In some embodiments, the track system **1201** is an alignment system. In other embodiments, other alignment system can be used, such as first track **1208**, second track **1210**, or longitudinal rails, individual aligners for the respective solar panels that work collectively to alignment to solar panels, etc.
(88) Turning ahead in the drawings, FIG. **13** illustrates the torque tube support structure **1202** and the hard stops **1226** of the jig **1200** (FIG. **12**). The torque tube support structure **1202** can include stanchions **1300** that are configured to hold the torque tube **1206** in the fixed position (e.g., a fixed height). The stanchions **1300** can include saddles **1302** that are configured to receive the torque tube **1206**. In some embodiments, the stanchions **1300** can include a tube **1304** and a rod **1306** positioned within the tube **1304**. In some embodiments, the saddle **1302** is coupled to the rod **1306** and the saddle **1302** is positioned external to the tube **1304**.
(89) In some embodiments, a spacer rod **1308** passes through each of the stanchions **1300** and the spacer rod **1308** includes spacers **1310** positioned along a length of the spacer rod **1308**. The spacer rod **1308** can include enough spacers (e.g., **1310**) to separate each of the solar panels **1204** to be affixed to torque tube **1206**. For example, when the jig (**1200**) is configured to affix up to twelve solar panels (e.g., **1204**) to a torque tube (e.g., **1206**), the spacer rod can include eleven spacers (e.g., **1310**). In some embodiments, the spacers **1310** are sized based on a specification for a particular solar panel assembly. That is, the spacers **1310** are sized to space the solar panels **1204** apart a specified distance so that the torque tube **1206** can be coupled to the solar panels **1204** to form a solar panel assembly. For example, the spacers can have a thickness of approximately ⅜

inch (0.95 centimeters (cm)), or another suitable thickness. In some embodiments, such as those shown in FIG. **13**, the spacers can be discs. In other embodiments, other suitable shapes can be used.
(90) In some embodiments, a handle rod **1312** passes through each of the stanchions **1300** and is operable to pivot away the stanchions **1300** from the track system **1201** to assist with removing the solar panel assembly once it is assembled. In some embodiments, after the solar panels **1204** have been affixed to the torque tube **1206**, an operator can utilize the handle rod **1312** to facilitate pulling the solar panel assembly away from the track system **1201** so that the solar panel assembly can be moved to be loaded onto a cassette or positioned on pile caps for installation. In some embodiments, automated controls can utilize the handle rod **1312** to facilitate pulling the solar panel assembly away from the track system **1201** so that the solar panel assembly can be moved to be loaded onto a cassette or positioned on pile caps for installation.
(91) In some embodiments, the hard stops **1226** can be positioned at an end of the track system **1201**. In some embodiments, the hard stops **1226** are configurable to abut ends of the various different torque tubes (e.g., **1206**) at different positions. As explained above, a solar panel assembly specification can specify that a torque tube of a solar panel assembly intended for a certain installation position has a certain amount of overhang at the end of the solar panel assembly beyond the end of the solar panels in order to be coupled with a neighboring solar panel assembly. In some embodiments, one of the hard stops **1226** can be selected and utilized to abut an end of the torque tube **1206** to account for the overhang. In some embodiments, a subsequent solar panel assembly can involve another amount of overhang different from the previous overhang. Accordingly, another one of the hard stops **1226** can be utilized to abut an end of a subsequent torque tube to account for the overhang. In many embodiments, the hard stops **1226** can be similar or identical to the hard stops **1116** (FIG. **11**), and the selected one of the hard stops **1116** (FIG. **11**) used for a particular torque tube on torque tube load rack **1114** (FIG. **11**) can correspond to the selected one of the hard stops **1226** used when the same torque tube is moved to torque tube support structure **1202**.
(92) Turning ahead in the FIG. **14** the panel aligner **1400** of the jig **1200** (FIG. **12**). The panel aligner **1400** is merely exemplary, and embodiments of the panel aligner are not limited to embodiments presented herein. The panel aligner can be employed in many different embodiments or examples not specifically depicted or described herein. In some embodiments, the panel aligner **1400** can be configured to rotationally align the solar panels **1204** with each other while the solar panels **1204** are in the track system **1201**. In some embodiments, first track **1208** can be aligned with respect to base **1100** using leveling screws **1410**.
(93) In some embodiments, the panel aligner **1400** can include a guide **1401**, a bladder **1402**, and/or a biasing element **1404**. In some embodiments, the solar panels **1204** can be operated along the automated conveyor **1212** until the solar panels **1204** are in a desired position. In some embodiments, when the solar panels **1204** are in the desired position, the bladder **1402** fills with a fluid (e.g., air, liquid, mechanical actuator, etc.) to cause the guide **1401** to abut and push the solar panels **1204**, to rotationally align the solar panels **1204** with each other. For example, the bottom edges of the solar panels **1204** can be aligned with each other. In some embodiments, the solar panels **1204** can be affixed to the torque tube **1206** (FIG. **12**) while the guide **1401** is abutting the solar panels **1204** to ensure proper alignment. In some embodiments, the bladder **1402** can release the fluid and the biasing element **1404** (e.g., a compression spring) biases the guide **1401** away from the solar panels **1204**. In some embodiments, the solar panels **1204** can be affixed to the torque tube **1206** (FIG. **6**) after the guide has retracted away from the solar panels **1204**. In such embodiments, the solar panels **1204** can remain in a rotationally aligned position after being aligned by panel aligned **1400**.
(94) Turning ahead in the drawings, FIG. **15** illustrates the compressing actuator **1224** of the jig **1200** (FIG. **12**). The compressing actuator **1224** is merely exemplary, and embodiments of the

compressing actuator are not limited to embodiments presented herein. The compressing actuator can be employed in many different embodiments or examples not specifically depicted or described herein. In some embodiments, the compressing actuator **1224** can be configured to narrow a respective spacing between each of the solar panels **1204** on the track system **1201**.

(95) In some embodiments, the compressing actuator **1224** can include a motor **1500**, a lead screw **1502** coupled to the motor **1500**, and an end guide **1504** coupled to the lead screw **1502**. In some embodiments, the motor **1500** is configured to operate the lead screw **1502** to extend the end guide **1504** to abut an end of one of the solar panels **1204** and move the solar panels **1204** along a direction of a longitudinal length of the lead screw **1502** to narrow the respective spacing between each of the solar panels **1204** on the track system **1201**. In some embodiments, the motor **1500** extends the end guide **1504** to narrow the spacing between the solar panels until each side of a respective spacer **1310** is abutting a solar panel **1204** (as illustrated in FIG. **15**). In some embodiments, the motor **1500** operates the end guide **1504** until the solar panels **1204** are aligned based on a solar panel assembly specification. For example, a spacing between each of the solar panels **1204** can be narrowed from approximately 1½ inch (3.81 cm) to approximately ⅜ inch (0.95 cm), or another suitable spacing. In some embodiments, the compressing actuator **1224** may not have the motor **1500**. In such an embodiment, an operator can physically maneuver the end guide **1504** until the solar panels **1204** are aligned based on a solar panel assembly specification.

(96) Turning ahead in the drawings, FIG. **16** illustrates a lengthwise cross-section of the spacer rod **1308**. In some embodiments, the spacer rod **1308** can include a rigid sleeve **1600** and a compression spring **1602** positioned along the spacer rod **1308** in between each of the spacers **1310**. In some embodiments, the rigid sleeves **1600** are sized to be less than a width of each of the solar panels **1204** (FIG. **12**). In some embodiments, the compression springs **1602** are sized to be more than a width of each of the solar panels when the compression springs **1602** are uncompressed.

(97) In some embodiments, a solar panel assembly configuration does not include a full set of solar panels (e.g., a space is open between respective spacers **1310**). In such embodiments, the compressing actuator **1224** (FIGS. **12**, **15**) compresses the solar panels **1204** (FIG. **12**), but in locations where a solar panel **1204** (FIG. **12**) is not positioned in between respective spacers **1310**, the compression springs **1602** are compressed and the rigid sleeves **1600** engage the spacers **1310** in a manner similar to a solar panel **1204** (FIG. **12**), thus forming a space between the solar panels **1204** (FIG. **12**) the total thickness of two of the spacers **1310** and the rigid sleeve **1600**. The rigid sleeves **1600** can thus mitigate the spacers **1310** from engaging and reduce the time associated with assembling a solar panel assembly as the compressing actuator **1224** (FIG. **12**, **15**) does not compress an entire length of a solar panel **1204** (FIG. **12**). That is, the rigid sleeves **1600** can reduce the distance the compressing actuator **1224** (FIGS. **12**, **15**) operates to narrow the space between solar panels **1204** (FIG. **12**). In some embodiments, the compression springs **1602** can bias the spacers **1310** outward beyond a width of the solar panels **1204** (FIG. **12**) so that the solar panels **1204** can be readily inserted in a respective space **1604** to be aligned for coupling with the torque tube **1206** (FIG. **12**). In some embodiments, the rigid sleeves **11200** can be configured to be installed and/or removed on the spacer rod **1308** without removing the spacers **1310** from the spacer rod **1308**, In some embodiments, various different size of rigid sleeves **11200** can be used, such as to provide various different spaces between solar panels **1204** (FIG. **12**) when there is no solar panel used between a pair of the spacers **1310**.

(98) Returning to FIG. **12**, in some embodiments, the first assembly station **526** can include a conveyor track **1228** that is parallel to the track system **1201** and includes a load balancer **1230** and a retaining tool **1232**. In some embodiments, the conveyor track **1228** can be utilized to maneuver solar panels **1204** along a length of the track system **1201** and/or to move solar panel assemblies from the first assembly station **526** to the cassette. In some embodiments, the first assembly station **526** can include a second monorail **1234** that is perpendicular to the track system **1201** and is

configured to move the respective torque tubes **1206** from the torque tube load rack **1114** to the torque tube support structure **1202**. In some embodiments, the second monorail includes a load balancer **1236** and a retention tool **1238**.

(99) In some embodiments, the second assembly station **528** (FIG. **5**A) can be similar or identical to the first assembly station **526**, and various elements of the second assembly station **528** (FIG. **5**A) can be similar or identical to various elements of the first assembly station **526**. In some embodiments, the second assembly station **528** (FIG. **5**A) can operates in a similar manner or identical to the first assembly station **526**. In some embodiments, the assembly stations **526**, **528** can assemble solar panel assemblies concurrently by receiving torque tubes from a torque tube sorting area. The assembly stations **526**, **528** each affix the solar panels to the torque tube to form the solar panel assembly. In some embodiments, the wire harness is affixed after the solar panel assembly is transferred to the cassette, while the next solar panel assembly to be loaded on the cassette is being assembled at the assembly station (e.g., **526**, **528**),

(100) FIG. **17** illustrates a perspective view of a strip station **1700**, which can be included in another embodiment of the mobile solar panel assembly facility **500** (FIG. **5**A). In the illustrated embodiment of FIG. **12**, the pallet stripping area **564** of solar panel assembly facility includes the strip station **1700**. The strip station **1700** can include a conveyor platform **1702**, a lift **1704**, a rotating drum **1706**, and/or conveyors **1708**.

(101) During operation, solar panel packages **570** are unloaded from a vehicle at the loading area and positioned on the pallet conveyor. The pallet conveyor operates to move the solar panel packages **570** toward the pallet stripping area **564**. At the pallet stripping area **564**, the solar panel packages **570** are unpacked and the waste is unloaded at the waste station. However, the solar panel packages **570** may have been unloaded from the vehicle in an orientation that is different from how they are to be loaded at the accumulation area (e.g., **566** (FIG. **5**)). The strip station **1700** can move and rotate the solar panels **574** so that the solar panels are in a proper orientation for the accumulation area, thereby reducing labor and increasing efficiency at the mobile solar panel assembly facility **500**. The solar panels **574** are loaded onto the conveyor platform **1702** and the lift **1704** lifts the conveyor platform **1702** to a height that aligns the solar panels **574** with the conveyors **1708**. The conveyors **1708** are positioned on a top, bottom, left and right side of the inside of the rotating drum **1706** to prevent or limit movement of the solar panels **574**. The conveyors **1708** receive the solar panels **574** from the conveyor platform **1702**. Based on the specific orientation expected for a solar panel assembly, the rotating drum **1706** will rotate a certain number of degrees to ensure the solar panels **574** are in the proper orientation. In many embodiments, the solar panels **574** can be loaded into the rotating drum **1706** as a set of solar panels, such as all of the solar panels included in one package of solar panels.

(102) Turning to FIG. **18**, a side view of the rotating drum **1706** is illustrated. FIG. **19** illustrates a perspective view of the rotating drum **1706** and a de-stacker **1812**. FIG. **20** illustrates an end view of the rotating drum and the de-stacker **1812**. As shown in FIG. **18**, the rotating drum **1706** can include motors **1810** that operate to rotate the rotating drum **1706** based on the desired specifications of a solar panel assembly. As shown in FIGS. **18-19**, the solar panels **574** that were loaded in the rotating drum **1706** FIG. **17** have been rotated 90 degrees to an upright position so that they can be unloaded with the de-stacker **1812**, which can include rolling rods **1814** to allow the solar panels **574** to be transferred onto the conveyor track **1228** (e.g., which can be similar or identical to conveyor **578** (FIG. **5**) of the accumulation area **566** (FIG. **5**)). The de-stacker **1812** can move the solar panels **574** into the accumulation area to be fed toward the first assembly station **526** (FIG. **5**A), where the solar panels can be aligned and coupled together (discussed in more detail above) on a torque tube, after which the solar panel assembly is loaded onto a cassette and transferred to an installation site. In some embodiments, the solar panels **574** moved from the de-stacker **1812** to the accumulation area **566** (FIG. **5**) to maintain the conveyor track **1228** at full capacity with solar panels. For example, each solar panel from the solar panels **574** is loaded

individually onto the conveyor track **1228** from the de-stacker **1812**. The conveyor track **1228** can move the solar panels along to the first assembly station **526** (FIGS. **5**A-**5**B), and additional solar panels **574** can be fed into the conveyor track **1228** from the de-stacker **1812** to fill the space of the solar panels **574** that have been positioned in first assembly station **526**.
(103) Turning to FIG. **21**, FIG. **21** illustrates a method **2100** of loading a cassette, according to another embodiment. Method **2100** is merely exemplary and is not limited to the embodiments presented herein. Method **2100** can be employed in many different embodiments or examples not specifically depicted or described herein. In some embodiments, the procedures, the processes, and/or the activities of method **2100** can be performed in the order presented. In other embodiments, the procedures, the processes, and/or the activities of method **2100** can be performed in any suitable order. In still other embodiments, one or more of the procedures, the processes, and/or the activities of method **2100** can be combined or skipped.
(104) The embodiments disclosed herein can advantageously allow for rapid loading of solar panel assemblies on a cassette for loading onto a vehicle, so the vehicle can readily install the solar panel assemblies at an installation site. By pre-loading solar panel assemblies, the solar panel assemblies can be rapidly installed once at the installation site. In some embodiments, method **2100** can include an activity **2105** of transferring a cassette from a first vehicle to an unload queue of a mobile solar panel assembly facility. The cassette comprises slots configured to hold solar panel assembles, and the slots of the cassette are empty when the cassette is transferred to the unload queue. In some embodiments, method **2100** can include an activity **2110** of transferring the cassette from the unload queue to a load queue of the mobile solar panel assembly facility. In some embodiments, method **2100** can include an activity **2115** of loading solar panel assemblies from the mobile solar panel assembly facility into the slots of the cassette while the cassette is in a fill position of the load queue. In some embodiments, method **2100** can include an activity **2120** of transferring the cassette with the slots filled with the solar panel assemblies from the load queue to a second vehicle to transfer the mobile solar panel assemblies to an installation site.
(105) Turning to FIG. **22**, FIG. **22** illustrates a method **2200** of installing solar panel assemblies, according to another embodiment. Method **2200** is merely exemplary and is not limited to the embodiments presented herein. Method **2200** can be employed in many different embodiments or examples not specifically depicted or described herein. In some embodiments, the procedures, the processes, and/or the activities of method **2200** can be performed in the order presented. In other embodiments, the procedures, the processes, and/or the activities of method **2200** can be performed in any suitable order. In still other embodiments, one or more of the procedures, the processes, and/or the activities of method **2200** can be combined or skipped.
(106) The embodiments disclosed herein can advantageously allow for rapid loading and unloading of solar panel assemblies, so the vehicle can readily install the solar panel assemblies at an installation site. By pre-loading solar panel assemblies, the solar panel assemblies can be rapidly installed once at the installation site. In some embodiments, method **2200** can include an activity **2205** of loading solar panel assemblies onto a vehicle. In some embodiments, method **2200** can include an activity **2210** of transporting the solar panel assemblies on the vehicle to a pile row of a solar project installation site. In some embodiments, method **2200** can include an activity **2215** of individually unloading the solar panel assemblies along the pile row at the installation site to facilitate installation of the solar panel assemblies on the pile row.
(107) FIG. **23** illustrates a perspective view of an assembly station **2300**, according to another embodiment. Assembly station **2300** can be similar to assembly station **526** (FIG. **5**A), and various elements of assembly station **2300** can be similar or identical to various elements of assembly station **526** (FIG. **5**A). The assembly station **526** can include the torque tube load rack **1114**. In some embodiments, the torque tube load rack **1114** can include the hard stops **1116** (FIG. **11**) or another linear positioner, and the stanchions **1118**, which can be used to hold a torque tube (e.g., **1120**). In the illustrated embodiment of FIG. **23**, the stanchions **1118** are hydraulic stanchions,

which can be adjusted to various different heights using a foot-activated actuator. In some embodiments, the stanchions **1118** are in a fixed position and are manually or automatically adjusted based on the specifications of a solar panel assembly. The stanchions **1118** can be configured to receive a torque tube **1120**. The stanchions **118** include saddles **2301**, and, the saddles **2301** include rollers **2302** to allow the rollers **2302** to allow the torque tube **1120** to readily move along the saddles **2301**, which can involve sliding the torque tube within the brackets manually or automatically.

(108) In the illustrated embodiment, a bracket loader system **2304** is positioned adjacent to the torque tube load rack **1114**. The bracket loader system **2304** includes a platform **2306**, rails **2308**, and bracket loaders **2310** coupled to the rails **2308**. The bracket loaders **2310** can be moved along the length of the rails based on the specifications of a particular solar panel assembly. The bracket loaders **2310** can include aligners **2312** that receive a bracket from the main body **2314** of the bracket loader **2310** to align the bracket with the saddle **2301** so that a torque tube **1120** can be slide along the saddles **2301**, through the brackets held by the aligners **2312**. For example, an alignment device can align ends of the various different torque tubes (e.g., **1120**) at various different positions, as described above, and the torque tube can be moved along the torque tube load rack **1114** to the proper position. In some embodiments, a solar panel assembly specification can specify that a torque tube of a solar panel assembly intended for a certain installation position has a certain amount of overhang at the end of the solar panel assembly beyond the end of the solar panels in order to be coupled with a neighboring solar panel assembly. In various embodiments, the bracket loaders **2310** can be positioned along the rails **2308** relative to a saddle **2301** and position a bracket (e.g., bracket **1122** (FIG. **11**)) in the aligner **2312** and the torque tube **1120** can be loaded into torque tube load rack **1114** and passed through the brackets **1122** (FIG. **11**) in the aligners **2312**. In some embodiments, brackets **1122** (FIG. **11**) and bushings **1124** (FIG. **11**) can be coupled to the torque tube **1120** based on the specifications of the particular solar panel assembly. In some embodiments, the brackets **1122** (FIG. **11**) are positioned on the torque tube **1120** based on respective locations of corresponding solar panels, and the bushings **1124** (FIG. **11**) are positioned on the torque tube **1120** based on respective locations of corresponding pile caps at an installation site. In some embodiments, the positions of the brackets and/or the bushings can be specified in the solar panel assembly specification or from as-built pile locations. In some embodiments, the brackets and/or bushings, as well as other mechanical or electrical hardware, can be installed on the solar panel assembly on either side of the solar panel. For example, an operator can install brackets on one side of the solar panel (e.g., a side with a torque tube) and install other hardware (e.g., mechanical, electrical) on the opposite side of the solar panel. In some embodiments, the brackets and other hardware (e.g., mechanical, electrical) can be coupled to the solar panels via an autonomous robot based on the specifications of the solar panel assembly.

(109) Turning to FIG. **24**, an internal view of the bracket loader **2310**. The main body **2314** of the bracket loader **2310** can include a bracket conveyor **2316** that includes portions **2318** to receive brackets **1122**. During operation, the portions **2318** of the bracket conveyor **2316** are loaded with brackets **1122**. Prior to a torque tube being loaded in the torque tube load rack **1114**, the bracket loader **2310** moves along the rails **2308** to a specified position, the bracket conveyor **2316** operates to move the brackets **1122** toward the aligner **2312**, and a bracket **1122** falls into position in the aligner **2312** when it reaches an end **2320** of the bracket loader **2310** and slides along a track **2322**.

(110) FIG. **25** illustrates a method **2500** of assembling a solar panel assembly, according to another embodiment. Method **2500** is merely exemplary and is not limited to the embodiments presented herein. Method **2500** can be employed in many different embodiments or examples not specifically depicted or described herein. In some embodiments, the procedures, the processes, and/or the activities of method **2500** can be performed in the order presented. In other embodiments, the procedures, the processes, and/or the activities of method **2500** can be performed in any suitable order. In still other embodiments, one or more of the procedures, the processes, and/or the activities

of method **2500** can be combined or skipped. In many embodiments, method **2500** can be performed using a jig (e.g., **1200** (FIG. **12**)) and/or a mobile solar panel assembly facility (e.g., **500** (FIG. **5**A)).

(111) In some embodiments, method **2500** of assembling a solar panel assembly can include an activity **2505** of transferring respective solar panels (e.g., **574** (FIG. **5**A), **1204** (FIG. **12**)) from a solar panel loading and sorting station (e.g., **560**, **580** (FIG. **5**A)) to an alignment system, such as track system (e.g., **1201** (FIG. **12**)) of a mobile solar panel assembly facility (e.g., **500** (FIG. **5**A)). In some embodiments, when transferring the respective solar panels to the alignment system (e.g., track system), the method further can include an activity **2510** of positioning the respective solar panels in a first track (e.g., **1208** (FIG. **12**)) and a second track (e.g., **1210** (FIG. **12**)).

(112) In some embodiments, the method can include an activity **2515** of transferring a respective torque tube from a torque tube queue or sorting area of the mobile solar panel assembly facility to a torque tube load rack (e.g., **1114** (FIG. **11**)). In some embodiments, when transferring the respective torque tube or sorting area to the torque tube assembly station the method further can include an activity **2520** of positioning the respective torque tube on respective stanchions of the torque tube load rack.

(113) In some embodiments, the method can include an activity **2525** of positioning brackets and/or bushings to the torque tube at the torque tube load rack. In some embodiments, when positioning or coupling brackets to the torque tube at the torque tube assembly station, the method further can include an activity **2530** of aligning the brackets on the torque tube based on a corresponding position of one of the respective solar panels. In some embodiments, the method can include an activity **2535** of moving the torque tube from the torque tube load rack to a torque tube support structure (e.g., **1202** (FIG. **12**)). In some embodiments, the method can include an activity **2540** of positioning the torque tube on a hard stop to align the torque tube with a respective position on the respective solar panels.

(114) In some embodiments, the method further can include an activity **2545** of compressing the respective solar panels until each of the respective solar panels abuts a respective spacer disc (e.g., **1310** (FIG. **13**)), which can involve narrowing a respective gap between each of the solar panels. In some embodiments, the method can include an activity **2550** of simultaneously aligning the respective solar panels at the alignment system (e.g., track system).

(115) In some embodiments, the method can include an activity **2555** of affixing the respective solar panels to the torque tube on the torque tube support station via the brackets to form a solar panel assembly. In some embodiments, the method can include an activity **2560** of removing the solar panel assembly from the alignment system (e.g., track system). In some embodiments, when removing the solar panel assembly from the alignment system (e.g., track system), the method can include an activity **2565** of moving the solar panel assembly in a first direction away from the alignment system (e.g., track system), and lifting the solar panel assembly via a work tool. In some embodiments, the method can include an activity **2570** of positioning the solar panel assembly on a cassette. In some embodiments, the solar panel assembly is a first solar panel assembly having a first number of solar panels and a first torque tube having a first length, and the method can include an activity **2575** of forming a second solar panel assembly with a second number of solar panels different than the first number of solar panels and a second torque tube having a second length different than the first length.

(116) The embodiments disclosed herein can advantageously allow for controlled installation of prefabricated solar panel assemblies of solar panels, trackers (torque tubes), and wire-harnesses, which are then placed with monorail load balancers onto a previously installed row of piles. These techniques eliminate the logistics of site-wide material pre-staging and multiple visits to every location by successive teams of craftsmen, quality control, and supervision. The operational methodology can involve the solar panel assembly installation vehicle picking up a full populated cassette from the mobile solar panel assembly facility **500** (FIG. **5**A), then transporting the full

cassette to the row of empty piles intending to receive completed assemblies. The assemblies are installed on the pile caps, aligned, torqued in place, with final inspection buyoff obtained before the solar panel assembly installation vehicle moves to next successive position. This process can thus create fully inspected and consistent quality assemblies successively until the solar panel assembly installation vehicle returns to the mobile solar panel assembly facility for material reload.
(117) The embodiments disclosed herein can advantageously allow for controlled installation of prefabricated solar panel assemblies of solar panels, trackers (torque tubes), and wire-harnesses, which are then placed with monorail load balancers onto a previously installed row of piles. These techniques eliminate the logistics of site-wide material pre-staging and multiple visits to every location by successive teams of craftsmen, quality control, and supervision. The operational methodology can involve the solar panel assembly installation vehicle **600** (FIG. **6**) picking up a full populated cassette from the mobile solar panel assembly facility **500** (FIGS. **5**A-**5**G), then transporting the full cassette to the row of empty piles intending to receive completed assemblies. The assemblies are installed on the pile caps, aligned, torqued in place, with final inspection buyoff obtained before the solar panel assembly installation vehicle **600** (FIG. **6**) moves to next successive position. This process can thus create fully inspected and consistent quality assemblies successively until the solar panel assembly installation vehicle **600** (FIG. **6**) returns to the mobile solar panel assembly facility **500** (FIGS. **5**A-**5**G) for material reload.
(118) Concurrently, while the vehicle is in transit and engaged in placing solar panel assemblies, other cassettes are populated with new assemblies for the next successive rows to be built. Serial activities are minimized improving cycle time by using multiple vehicles and maintaining production at a rate sufficient to keep cassettes populated and ready to move to vehicles when they return and dispense with an empty cassette. Overall, embodiments disclosed herein maintain assembly fabrication in one location and use vehicles and methods to install the assemblies at a rapid pace while maintaining verifiable installation quality.
(119) In some embodiments, the techniques described herein can beneficially reduce the amount of time used to install a solar project over multiple square miles by significant decreasing the time involved to install the solar panels in the solar project, while also significantly reducing the number of workers involved in the installation project. Meanwhile, the quality of installation provided by the techniques described herein is superior to that of conventional techniques.
(120) It should be recognized that numerous variations can be made to the above-described systems and methods without departing from the scope of the invention.
(121) While various novel features of the invention have been shown, described, and pointed out as applied to particular embodiments thereof, it should be understood that various omissions, substitutions and changes in the form and details of the systems and methods described and illustrated may be made by those skilled in the art without departing from the spirit of the invention. Amongst other things, the steps shown in the methods may be carried out in different orders in many cases, where such may be appropriate. Those skilled in the art will recognize, based on the above disclosure and an understanding therefrom of the teachings of the invention, that the particular hardware and devices that are part of the system described herein, and the general functionality provided by and incorporated therein, may vary in different embodiments of the invention. Accordingly, the particular system components are for illustrative purposes to facilitate a full and complete understanding and appreciation of the various aspects and functionality of particular embodiments of the invention, as realized in system and method embodiments thereof. Those skilled in the art will appreciate that the invention can be practiced in other than the described embodiments, which are presented for purposes of illustration and not limitation. For example, to one of ordinary skill in the art, it will be readily apparent that any element of FIGS. **1-25** may be modified, and that the foregoing discussion of certain of these embodiments does not necessarily represent a complete description of all possible embodiments.
(122) Replacement of one or more claimed elements constitutes reconstruction and not repair.

Additionally, benefits, other advantages, and solutions to problems have been described with regard to specific embodiments. The benefits, advantages, solutions to problems, and any element or elements that may cause any benefit, advantage, or solution to occur or become more pronounced, however, are not to be construed as critical, required, or essential features or elements of any or all of the claims, unless such benefits, advantages, solutions, or elements are stated in such claim.
(123) Moreover, embodiments and limitations disclosed herein are not dedicated to the public under the doctrine of dedication if the embodiments and/or limitations: (1) are not expressly claimed in the claims; and (2) are or are potentially equivalents of express elements and/or limitations in the claims under the doctrine of equivalents.

Claims

1. An apparatus comprising: a base; and stanchion rows configured to hold solar panel assemblies, wherein each one of the stanchion rows is configured to hold a torque tube of a respective solar panel assembly of the solar panel assemblies, and the each one of the stanchion rows comprises respective stanchions, wherein the stanchion rows are configured to hold the solar panel assemblies when lengths of torque tubes of the solar panel assemblies are different and when the lengths of the torque tubes are the same.
2. The apparatus of claim 1, wherein each of the respective stanchions of the each one of the stanchion rows comprises a respective saddle configured to receive the torque tube of the respective solar panel assembly held by the each one of the stanchion rows.
3. The apparatus of claim 2, wherein: the each of the respective stanchions comprises a respective tube and a respective rod positioned in the respective tube; and the respective saddle of the each of the respective stanchions is coupled to the respective rod.
4. The apparatus of claim 3, wherein the respective saddle is positioned external to the respective tube.
5. The apparatus of claim 3, wherein the respective rod is coupled to an actuator on the base.
6. The apparatus of claim 5, wherein the actuator operates the respective rod to extend the respective saddle away from respective tube.
7. An apparatus comprising: a base; and stanchion rows configured to hold solar panel assemblies, wherein each one of the stanchion rows is configured to hold a torque tube of a respective solar panel assembly of the solar panel assemblies, and the each one of the stanchion rows comprises respective stanchions, wherein: each of the respective stanchions of the each one of the stanchion rows comprises a respective saddle configured to receive the torque tube of the respective solar panel assembly held by the each one of the stanchion rows; the base comprises one or more respective recesses corresponding to the each one of the stanchion rows; and the one or more respective recesses are configured to receive a portion of one or more solar panels of the respective solar panel assembly held by the each one of the stanchion rows when the torque tube of the respective solar panel assembly is positioned in the respective saddles of the respective stanchions of the each one of the stanchion rows.
8. The apparatus of claim 1, wherein the base comprises one or more apertures configured to receive one or more tools or conveyors of a work vehicle.
9. The apparatus of claim 1, wherein the base comprises a first set of wheels and a second set of wheels.
10. The apparatus of claim 9, wherein: the first set of wheels is positioned in a first direction; and the second set of wheels is positioned in a second direction different from the first direction.
11. The apparatus of claim 9, wherein: the base comprises drive motors configured to drive the first and second sets of wheels; and the base comprises a battery configured to power the drive motors.
12. The apparatus of claim 1, wherein a quantity of the respective stanchions of the each one of the stanchion rows is at least three.

13. The apparatus of claim 1, wherein one or more of the respective stanchions of the each one of the stanchion rows are configured to rotate between an upright position and a lowered position.
14. Apparatus comprising: a base; and stanchion rows configured to hold solar panel assemblies, wherein each one of the stanchion rows is configured to hold a torque tube of a respective solar panel assembly of the solar panel assemblies, and the each one of the stanchion rows comprises respective stanchions, wherein: one or more of the respective stanchions of the each one of the stanchion rows are configured to rotate between an upright position and a lowered position; and the one or more of the respective stanchions are spring-biased in the upright position.
15. The apparatus of claim 1, wherein a quantity of the stanchion rows is at least six.
16. The apparatus of claim 1, wherein the solar panel assemblies held by the stanchion rows are configured to extend across a full block row of a solar project.
17. The apparatus of claim 1, wherein the solar panel assemblies configured to be held by the stanchion rows collectively comprise up to approximately 144 solar panels.
18. A method comprising: transferring a cassette from a first vehicle to an unload queue of a mobile solar panel assembly facility, wherein the cassette comprises slots configured to hold solar panel assemblies, and the slots of the cassette are empty when the cassette is transferred to the unload queue; transferring the cassette from the unload queue to a load queue of the mobile solar panel assembly facility; while the cassette is in a fill position of the load queue, loading solar panel assemblies from the mobile solar panel assembly facility into the slots of the cassette; transferring the cassette with the slots filled with the solar panel assemblies from the load queue to a second vehicle to transfer the solar panel assemblies to an installation site; and removing the solar panel assemblies from the cassette when the second vehicle arrives at the installation site, comprising individually lifting each respective one of the solar panel assemblies and transferring the each respective one of the solar panel assemblies in a first widthwise direction to a position above a respective installation position along a pile row of the installation site.
19. A method comprising: transferring a cassette from a first vehicle to an unload queue of a mobile solar panel assembly facility, wherein the cassette comprises slots configured to hold solar panel assemblies, and the slots of the cassette are empty when the cassette is transferred to the unload queue; transferring the cassette from the unload queue to a load queue of the mobile solar panel assembly facility; while the cassette is in a fill position of the load queue, loading solar panel assemblies from the mobile solar panel assembly facility into the slots of the cassette; and transferring the cassette with the slots filled with the solar panel assemblies from the load queue to a second vehicle to transfer the solar panel assemblies to an installation site, wherein loading the solar panel assemblies into the slots of the cassette further comprises, for each one of the solar panel assemblies, coupling a respective wire harness to the each one of the solar panel assemblies.
20. A method comprising; transferring a cassette from a first vehicle to an unload queue of a mobile solar panel assembly facility, wherein the cassette comprises slots configured to hold solar panel assemblies, and the slots of the cassette are empty when the cassette is transferred to the unload queue; transferring the cassette from the unload queue to a load queue of the mobile solar panel assembly facility; while the cassette is in a fill position of the load queue, loading solar panel assemblies from the mobile solar panel assembly facility into the slots of the cassette; and transferring the cassette with the slots filled with the solar panel assemblies from the load queue to a second vehicle to transfer the solar panel assemblies to an installation site, wherein loading the solar panel assemblies into the slots of the cassette further comprises, for each one of the solar panel assemblies, indexing a position of the cassette to align a respective next empty one of the slots with an assembling position of the mobile solar panel assembly facility to facilitate loading of the each one of the solar panel assemblies from the assembling position into the respective next empty one of the slots.
21. The method of claim 20, wherein loading the solar panel assemblies into the slots of the cassette further comprises, for the each one of the solar panel assemblies, raising respective

stanchions at the respective next empty one of the slots to hold the each one of the solar panel assemblies.
22. A method comprising: transferring a cassette from a first vehicle to an unload queue of a mobile solar panel assembly facility, wherein the cassette comprises slots configured to hold solar panel assemblies, and the slots of the cassette are empty when the cassette is transferred to the unload queue; transferring the cassette from the unload queue to a load queue of the mobile solar panel assembly facility: while the cassette is in a fill position of the load queue, loading solar panel assemblies from the mobile solar panel assembly facility into the slots of the cassette; and transferring the cassette with the slots filled with the solar panel assemblies from the load queue to a second vehicle to transfer the solar panel assemblies to an installation site, wherein: the unload queue comprises an unload conveyor configured to move the cassette within the unload queue; the load queue comprises a load conveyor configured to move the cassette within the load queue; and transferring the cassette from the unload queue to the load queue comprises using a transfer conveyor to move the cassette.
23. A method comprising: transferring a cassette from a first vehicle to an unload queue of a mobile solar panel assembly facility, wherein the cassette comprises slots configured to hold solar panel assemblies, and the slots of the cassette are empty when the cassette is transferred to the unload queue; transferring the cassette from the unload queue to a load queue of the mobile solar panel assembly facility; while the cassette is in a fill position of the load queue, loading solar panel assemblies from the mobile solar panel assembly facility into the slots of the cassette; and transferring the cassette with the slots filled with the solar panel assemblies from the load queue to a second vehicle to transfer the solar panel assemblies to an installation site, wherein the cassette comprises drive motors configured to transfer the cassette from the first vehicle to the unload queue, from the unload queue to the load queue, and from the load queue to the second vehicle.
24. A method comprising: transferring a cassette from a first vehicle to an unload queue of a mobile solar panel assembly facility, wherein the cassette comprises slots configured to hold solar panel assemblies, and the slots of the cassette are empty when the cassette is transferred to the unload queue; transferring the cassette from the unload queue to a load queue of the mobile solar panel assembly facility; while the cassette is in a fill position of the load queue, loading solar panel assemblies from the mobile solar panel assembly facility into the slots of the cassette; and transferring the cassette with the slots filled with the solar panel assemblies from the load queue to a second vehicle to transfer the solar panel assemblies to an installation site, wherein: the cassette moves in a first widthwise direction within the unload queue; the cassette moves in a second widthwise direction with the load queue; the second widthwise direction is opposite the first widthwise direction; and the cassette moves in a lengthwise direction when being transferred from the unload queue to the load queue.
25. The method of claim 18, wherein the first vehicle is different from the second vehicle.
26. The method of claim 18, further comprising lowering the each respective one of the solar panel assemblies onto respective pile caps at the respective installation position.
27. The method of claim 26, further comprising installing a coupler on a first torque tube of a first one of the solar panel assemblies to couple a second torque tube of a second one of the solar panel assemblies to the first torque tube.
28. The method of claim 27, further comprising moving the second one of the solar panel assemblies in a lengthwise direction toward the first one of the solar panel assemblies to position the second torque tube within the coupler and couple the second torque tube to the first torque tube.
29. The method of claim 18, further comprising driving the second vehicle forward to a respective subsequent one of the respective installation positions along the pile row between removing each respective one of the solar panel assemblies.
30. The method of claim 18, further comprising moving the cassette on the second vehicle in a second widthwise direction opposite the first widthwise direction to facilitate removing the solar

panel assemblies from the cassette.
31. The method of claim 18, wherein the solar panel assemblies are assembled at the mobile solar panel assembly facility.
32. The method of claim 18, wherein the solar panel assemblies loaded into the cassette are configured to extend across a full block row at the installation site.